

MSc. in Computing Practicum Approval Form

Project Title:	Structured Metadata Extraction from Consent Dialogues
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Topic:

The practicum focuses on the extraction of structured metadata from cookie consent dialogues and the building of an NLP model studying compliance with privacy regulations. This project uses both web scraping and methods of natural language processing to construct an automated system able to analyze the mechanism of consent, check compliance with privacy regulations, and bring in transparency in practices of data collection. The project will design and train, using transformer architecture, an NLP model that evaluates consent dialogues against the standards of GDPR, ISO 29184, and DPV-2, which will support the identification of compliance gaps and potential privacy issues.

The papers that we have read on this topic:

1. Data Privacy Vocabulary (DPV) - Version 2
2. Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence
3. Do Cookie Banners Respect my Choice? Measuring Legal Compliance of Banners from IAB Europe's Transparency and Consent Framework
4. Dark Patterns and the Legal Requirements of Consent Banners: An Interaction Criticism Perspective
5. A BERT-based Empirical Study of Privacy Policies' Compliance with GDPR

Relation to Existing Work:

Our project proposal extends existing research by combining the automated metadata extraction with NLP-based compliance analysis system. The Data Privacy Vocabulary (DPV) – version 2.0 paper provides the vocabulary for training our NLP model. The dark patterns research informs our model's feature engineering for detecting problematic practices in data collection. The IAB Europe's Transparency and Consent Framework compliance study offers

metrics for evaluating consent mechanisms. The BERT-based privacy policy study demonstrates the effectiveness of transformer models for privacy text analysis.

Our project innovates by:

- Developing an NLP model specifically for consent dialogue analysis
- Creating automated compliance checking using transformer architecture
- Implementing real-time analysis through browser extension
- Providing standardized metadata representation using DPV
- Contributing to privacy-focused NLP research

Research Questions:

1. How is metadata to be effectively extracted and represented from these various consent dialogues?
2. Which architecture is best in NLP for analyzing compliance in consent dialogues?
3. But how do we train them to spot compliance?
4. Can we create a standardized method to automate consent analysis?
5. How can we make compliance estimation transparent and explainable?

Exploration Methodology

Software/Programming Environment:

- Python with Selenium and BeautifulSoup for web scraping
- Transformers library (Hugging Face) for NLP model development
- PyTorch for deep learning implementation
- Pandas and scikit-learn for data processing

Coding/Development:

- Development of scraping scripts
- Implementation of transformer-based NLP model
- Creation of browser extension
- Development of compliance checking algorithms

Data Sources:

- Scraped consent dialogues
- Annotated compliance datasets
- Privacy regulation documents
- DPV-2 vocabulary

Data Availability:

Will be collected through:

- Web scraping
- Manual annotation
- Synthetic data generation
- Public privacy datasets

Experiments:

- Model training and validation
- Compliance analysis accuracy
- Pattern recognition testing
- Cross-validation with human experts

Expected Output:

- Trained NLP model
- Compliance analysis system
- Structured metadata representations
- Browser extension for real-time analysis

Evaluation:

Results will be evaluated through:

- Model performance metrics
- Compliance accuracy validation
- Expert review of findings
- User testing of browser extension